

Pier Two: Unleashing Web3 opportunities with innovation built on Google Cloud

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ABOUT PIER TWO

Pier Two built an Ethereum staking on Google Cloud enabled more than a million on-chain transactions, and computing costs to help drive collaboration in Web3 innovation.



About Pier Two

Pier Two is a global leader in Web3 innovation, enabling institutions to obtain yield through Ethereum staking without the burden of developing and maintaining blockchain infrastructure. Under the motto "*Your Keys, Your ETH, Always*," the non-custodial platform offers enterprise-grade infrastructure that delivers excellent uptime, supporting 1,000+ data feeds over 12 blockchains and facilitating more than 35 million transactions, since 2018. The Australian company received a grant from the Ethereum Foundation to develop an Ethereum

Google Cloud results

- Facilitates participation in Web3 solution building through 54.5% reduction cloud computing costs
- Enables limitless Web3 infrastructure growth through seamless 15,000% rise in memory on Compute Engine
- Supports agile and resilient blockchain validation and data services over 150 VM instances

17,000
increased
vCPU
solution

Light Client in C# as a potential alternative solution to interact with Ethereum, helping to support decentralization of the network.

Industries: Technology

Location: Australia

(<https://cloud.google.com/kubernetes-engine/>)

Google Cloud (<https://cloud.google.com/>)

Compute Engine
(<https://cloud.google.com/compute/>)

Cloud SQL (<https://cloud.google.com/sql/>)
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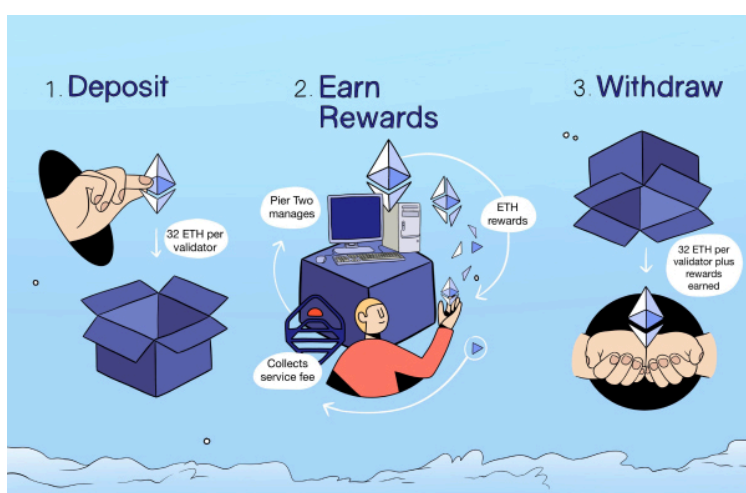
Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

- across 12
blockchains
- Enhances
Web3
innovation
mission
through open
partnership
in Blockchain
Node Engine
testing

Pier Two (<https://piertwo.com/>) is a Web3 innovator that seeks to open up the world of Ethereum staking to institutional clients, sparing them the burden of handling underlying infrastructure and maintaining security, while avoiding risks traditionally associated with decentralized finance (DeFi). In Ethereum staking, participants can lock up ETH to be rewarded with more ETH. This ETH is used by node operators to run validators. Each validator needs exactly 32 ETH to be activated, then actively securing the network.

Guided by the philosophy "Your Keys, Your ETH, Always," Pier Two is a non-custodial platform that enables customers to maintain control of their Ethereum assets, allowing them to receive and withdraw tokens at any time. Alongside control, Pier Two offers convenience. The only time ETH holders need to be involved in the transaction process is when they stake or unstake ETH, lowering barriers to entry.

By staking ETH on Pier Two, participants are able to put their Ethereum assets to work for the network. This means they can access rewards for staking that contributes to the growth and security of the Ethereum network. The opportunities Pier Two enables are based on the mechanism that grants a network participant the ability to act as a validator for the Ethereum blockchain by staking ETH.



The ability to stake and unstake on demand only started in 2023, following changes to the consensus mechanism of Ethereum from Proof-of-Work to Proof-of-Stake in "The Merge". This resulted in a dramatic reduction in energy consumption, of 99.9%.

As a Chainlink node operator, Pier Two sees its role as providing the "connective tissue" for global digital markets linking blockchain systems such as Ethereum to real-world customers and their data. To succeed, Pier Two elected to build its Chainlink node network on a powerful and nimble cloud environment. While many other Ethereum platforms were building on a rival public cloud, Pier Two selected [Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>) out

of a belief in the importance of decentralizing the spread of Chainlink node operators across cloud providers, ensuring redundancy in the emergent data systems powering Web3.

It was also impressed by the proactive, collaborative support offered by the Google Cloud team to optimize cloud solutions. The clincher was its range of user-friendly managed services which promised to significantly reduce the workload of its initial development team.

"We sought a cloud provider with proven and reliable managed services. Despite our small team size, we were able to build a product quickly thanks to the world-class support of Google Cloud," says Seamus McNab, Head of Infrastructure, Pier Two. "It was a fantastic choice. We discovered other platforms encountering network headaches on their provider which we just weren't on Google Cloud."

From Pier Two's beginnings in 2018, the collaborative relationship has evolved to the stage where Pier Two was invited to help test Google Cloud's **Blockchain Node Engine**

(<https://cloud.google.com/blog/products/infrastructure-modernization/introducing-blockchain-node-engine>)

(BNE) while in the preview stage. Pier Two has thus provided key insights that have played a direct role in improving the fully managed node-hosting solution for Web3 development ahead of general release. BNE is a fully managed dedicated node that removes the need for manually operating blockchain nodes, which are computers that contain a full copy of blockchain transaction history. "Google Cloud not

only makes sure that the nodes are running correctly, it also enables the best performance while making sure the software is up to date," says McNab. "It has been an exciting experience to work with some of the brightest minds at Google Cloud, and being able to share insights on Blockchain Node Engine from our years of experience optimizing our tech stack on Google Cloud."

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*—Seamus McNab, Head
of Infrastructure, Pier Two*

Driving unlimited Web3 transaction scale and agility with Compute Engine

Pier Two runs two main computing mechanisms. One is the Chainlink node network that automates the connection between a number of blockchains with customers. The other is the Ethereum validator infrastructure which spares customers the burden of managing a validator node to participate in the proof-of-stake process.

The platform maintains roughly 150 different virtual machine (VM) instances on Compute Engine (<https://cloud.google.com/compute>), mainly for the purpose of managing Web3 infrastructure for 12 different blockchains. "Since launching, we have grown from running Web3 infrastructure for one blockchain to operating across 12 different blockchains, and Compute Engine enables the agility and frictionless scaling to make that possible," says McNab. "The ability to quickly perform updates,

maintenance, and troubleshoot issues has allowed us to remain relevant and agile."

McNab says the ability to spin up, tear down, edit and migrate between VM instances within minutes on Compute Engine has been instrumental to Pier Two's success in offering instant, automated transactions on its platform, whatever the spikes in demand. Since 2018, Pier Two has enabled more than 35 million on-chain Ethereum transactions.

"Running 150 different instances, the ability to quickly change their size and performance is a major benefit," says McNab. "By easily swapping between machines and migrating workloads on Compute Engine, we have been able to reduce our cloud spend by 54.5% per month over the past year."

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He adds that an integrated, user-friendly Google Cloud ecosystem makes the management and maintenance of Pier Two infrastructure stack easier, averting the need to rely on outside software solutions that would complicate workloads and create potential security risks.

The frictionless scaling enabled by Google Cloud has underpinned impressive Pier Two growth. In its first two months, Pier Two managed two virtual CPUs (vCPUs) and seven gigabytes of memory. Now it handles 350 vCPU's and 1060 gigabytes of memory across its fleet of VMs. That translates into a 17,000% increase in vCPUs and 15,000% increase in memory across the Pier Two computing environment.

"The fact that we've been able to scale to the point of 17,000% increase in vCPUs, without any hiccups or downtime has been incredible," says McNab.

"Compute Engine brings us the agility and system resilience to achieve this scale, which simply wouldn't be sustainable in a data center environment."

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A future of dynamic Web3 solution building driven by creative partnerships

Thanks to a positive experience collaborating with Google Cloud on BNE testing, Pier Two sees promising potential for deploying the solution as it moves into general availability. McNab says the Pier Two team found the fully-managed node-hosting solution for Web3 development to be highly cost-effective, while providing a seamless automated experience.

Pier Two is also working on expansion of its ETH staking solution, with plans to explore the viability of other proof-of-stake blockchains. To facilitate this, it has been partnering with the Google Cloud team to optimize its Ethereum Infrastructure as Code (IaC) on Google Cloud. Through this symbiosis, it intends to create the technical foundation to deploy on new proof-of-stake platforms to grow its product suite and better serve customers.



Since 2021, Pier Two has supported Arbitrum, a layer two chain blockchain, and since 2022, maintains a privileged role in being one of only nine validators of Arbitrum along with ConsenSys, Ethereum Foundation, Offchain Labs, P2P, Quicknode and more.

Pier Two has also received a grant from the Ethereum Foundation itself to develop an Ethereum Light Client in C# designed to improve usability of the Ethereum ecosystem by eliminating the need to provision terabytes of storage for Web3 interactions. In this work-in-progress, Pier Two envisions potential uses for Cloud SQL (<https://cloud.google.com/sql/>) as a fully managed relational database and BigQuery (<https://cloud.google.com/bigquery/>) as an autoscaling data warehouse as project enablers.

"Our collaboration of Google Cloud is continuously evolving," says McNab. "With new managed services consistently being launched, combined with their powerful data analytics tools such as BigQuery, we envision an unlimited future of synergistic development for our Web3 solution."